ADVANCED MACHINE LEARNING · RESEARCH PAPER PROJECT 23

# Induction of Decision Trees

ID3 from scratch on 7 real datasets: information gain, depth control and noise

J. R. Quinlan (1986), Machine Learning 1(1): 81–106 · doi:10.1007/BF00116251

GROUP

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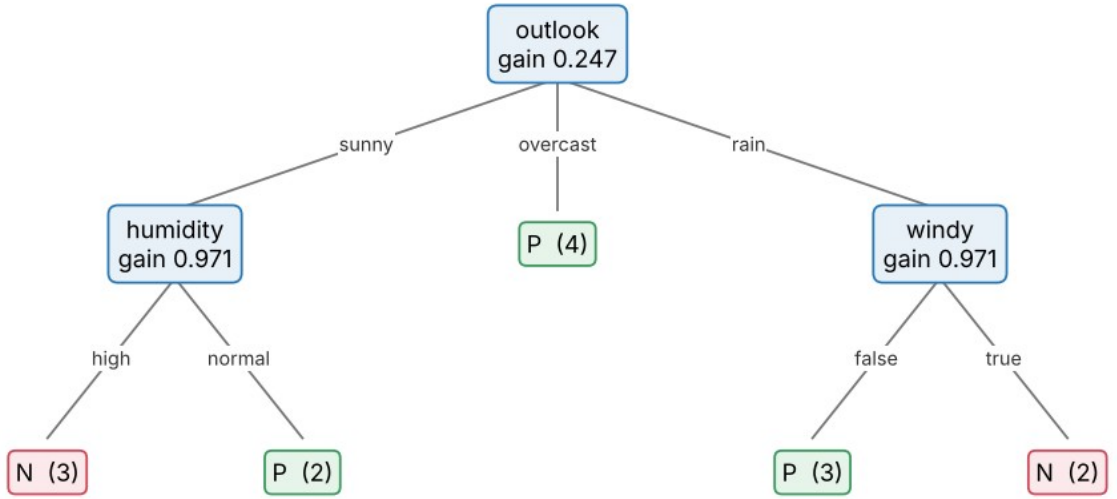
Code and data: github.com/Vinayak069/id3-decision-trees

MA30235 / MA60215 · Department of Mathematics, IIT Kharagpur · November 2026

# Learn a small tree that classifies objects from their attributes

| outlook | temperature | humidity | windy | class |
|---|---|---|---|---|
| sunny | hot | high | false | N |
| sunny | hot | high | true | N |
| overcast | hot | high | false | P |
| rain | mild | high | false | P |
| rain | cool | normal | false | P |
| rain | cool | normal | true | N |
| overcast | cool | normal | true | P |
| sunny | mild | high | false | N |
| sunny | cool | normal | false | P |
| rain | mild | normal | false | P |
| sunny | mild | normal | true | P |
| overcast | mild | high | true | P |
| overcast | hot | normal | false | P |
| rain | mild | high | true | N |

Table 1 of the paper: 14 Saturday mornings, P = suitable, N = not.




- **Induction task:** given objects described by categorical attributes and a class, build a rule that classifies unseen objects.
- **Top-down divide and conquer:** pick a test, split the objects, recurse until every subset is one class.
- **Occam's razor:** among trees that fit, prefer the simplest. ID3 gets there greedily, without searching all trees.

# ID3 splits on the attribute that carries most information about the class

$$I(C) = -\sum_k \frac{c_k}{|C|} \log_2 \frac{c_k}{|C|}$$

$$E(A) = \sum_{i=1}^{v} \frac{|C_i|}{|C|}\ I(C_i)$$

$$\text{gain}(A) = I(C) - E(A)\ =\ \text{MI}(A;\ \text{class})$$

$$IV(A) = -\sum_i \frac{|C_i|}{|C|} \log_2 \frac{|C_i|}{|C|}, \quad \text{ratio} = \frac{\text{gain}(A)}{IV(A)}$$

Worked root split: E(outlook) = 5/14 · 0.971 + 4/14 · 0 + 5/14 · 0.971 = 0.694
gain(outlook) = 0.940 − 0.694 = 0.246 bits (0.2467 unrounded)

| Attribute | gain (ours) | gain (paper) | IV | gain ratio | $\chi^2$ p-value |
|---|---|---|---|---|---|
| outlook | 0.247 | 0.246 | 1.577 | 0.156 | 0.170 |
| temperature | 0.029 | 0.029 | 1.557 | 0.019 | 0.752 |
| humidity | 0.152 | 0.151 | 1.000 | 0.152 | 0.094 |
| windy | 0.048 | 0.048 | 0.985 | 0.049 | 0.334 |

- **I(9, 5) = 0.940 bits;** outlook wins with 0.247 bits, so it becomes the root. Our values match the paper to its rounding.
- **Gain is mutual information** (paper footnote 3): ID3 asks which test most reduces uncertainty about the class.
- **Bias:** gain rewards attributes with many values. The gain ratio divides by IV(A) (Section 7): outlook 0.156 vs humidity 0.152.

METHOD

# We implemented every mechanism in the paper, then tested it

### Section 4 · Tree + window

- Recursive split on max gain; each attribute used once per path
- Empty branch → majority class of parent
- Windowing: grow on a sample, add misclassified objects, repeat

### Section 5 · Noise

- Impure leaves are labelled with the majority class
- $\chi^2$ test of independence: no split unless irrelevance is rejected ($\alpha$ = 1 % in the paper; we sweep $\alpha$)

### Section 6 · Unknown values

- Gain: spread unknowns over values $\propto |C_i|$, then discard them
- Classify: a token goes down every branch with weight $\text{ratio}_i$; class votes are summed

$$\chi^2 = \sum_i \sum_k \frac{(c_{ik} - c'_{ik})^2}{c'_{ik}}, \qquad c'_{ik} = c_k \frac{|C_i|}{|C|}$$

$$\text{split only if } \; P\left(\chi^2_{(v-1)(K-1)} > \chi^2\right) < 0.01$$

Pure Python / NumPy, about 500 lines (id3.py). sklearn is used only for the CART baseline.

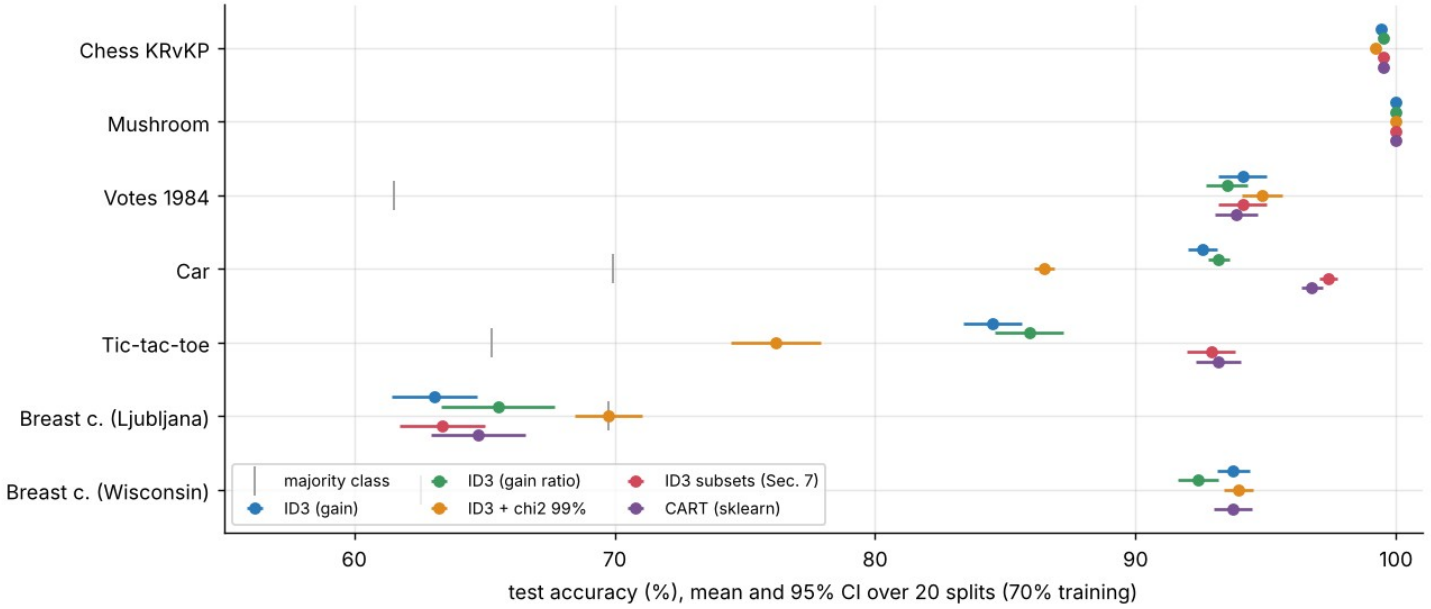
### Section 7 · Extensions

- Gain ratio among attributes with ≥ average gain
- Binary subset tests A ∈ S (ASSISTANT)
- Continuous attributes: threshold at midpoints

### Our additions

- max_depth for the depth study; K ≥ 2 classes
- 15 unit tests of the paper's claims
- Corrected paired t-test (Nadeau–Bengio)

# On 7 real datasets ID3 with subset tests matches CART

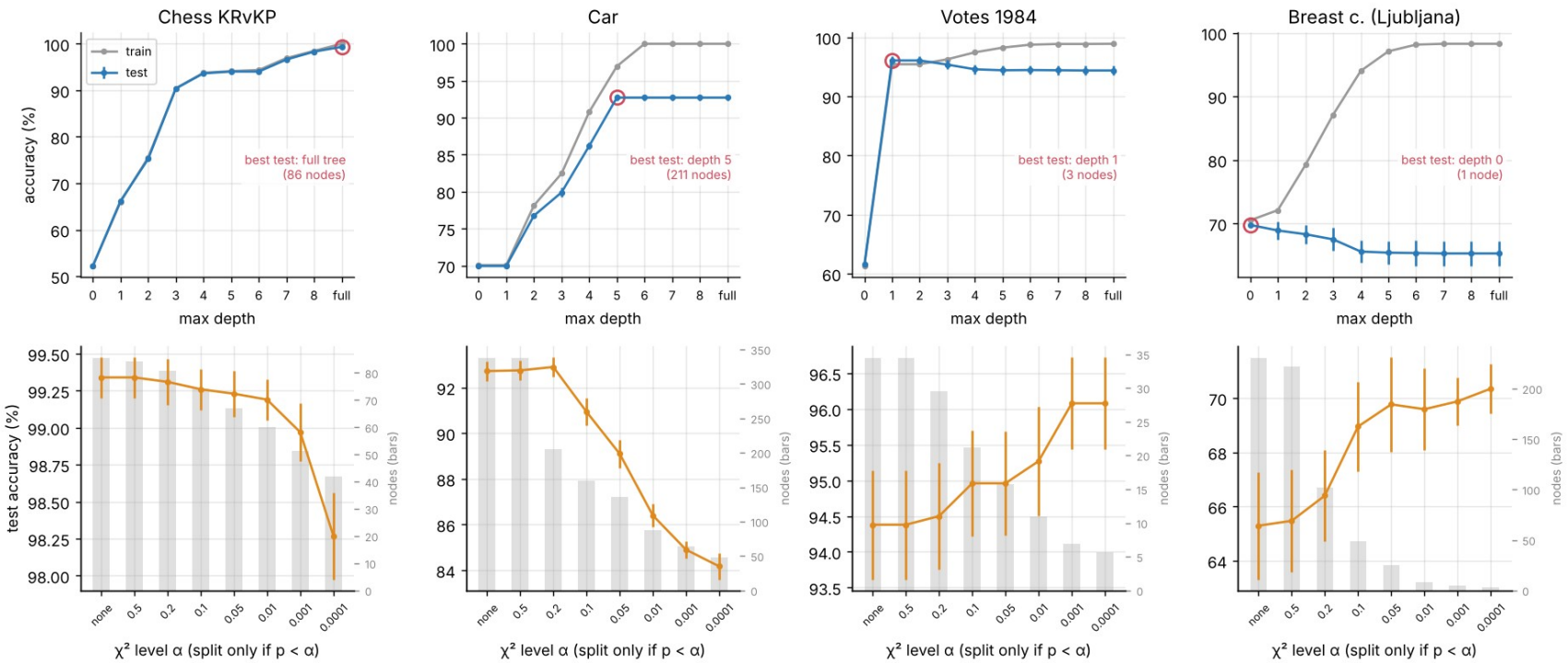



| Dataset | Objects | Attrs | Classes | '?' values |
|---|---|---|---|---|
| Chess KRvKP | 3 196 | 36 | 2 | 0 |
| Mushroom | 8 124 | 22 | 2 | 2480 |
| Votes 1984 | 435 | 16 | 2 | 392 |
| Car Evaluation | 1 728 | 6 | 4 | 0 |
| Tic-tac-toe | 958 | 9 | 2 | 0 |
| Breast c. Ljubljana | 286 | 9 | 2 | 9 |
| Breast c. Wisconsin | 569 | 30 | 2 | 0 |
| Sim: known tree | 2 000 | 10 | 2 | 0 |
| Sim + id_code | 600 | 11 | 2 | 0 |

UCI data (GitHub mirrors) + scikit-learn; simulated sets only for sanity checks.

- **Fragmentation is ID3's weak spot:** on Car and Tic-tac-toe, multi-way ID3 trails CART by 4–9 points. The paper's own binary subset tests close the gap (+4.8 and +8.4 points, 20/20 and 20/20 wins, $p < 0.001$).
- **Clean concepts (chess, mushroom):** all variants reach 99.4–100 %; differences are not significant.
- **Noisy medical data (Ljubljana):** the $\chi^2$ test gains 6.7 points over the full tree ($p = 0.04$) with 8 nodes instead of 224, but only reaches the majority-class rate: the attributes carry little signal.

# Depth control: shallow trees win on noisy data, full trees on clean data

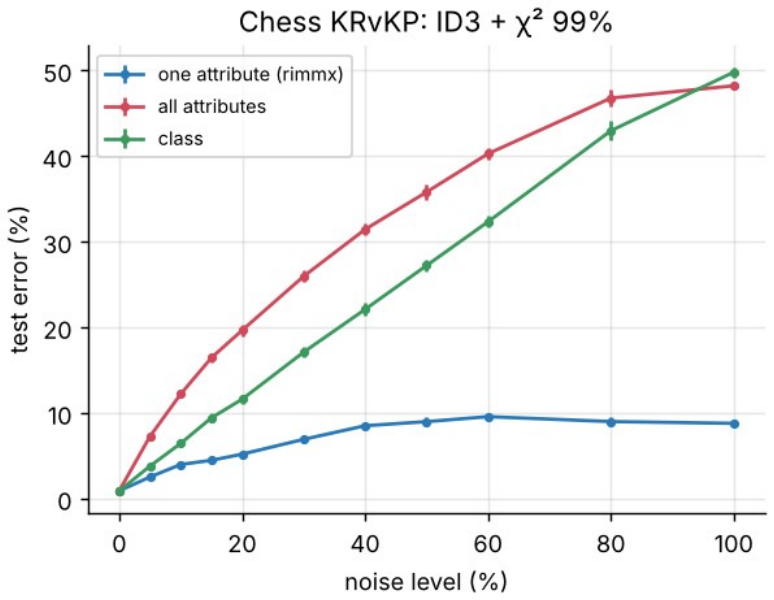


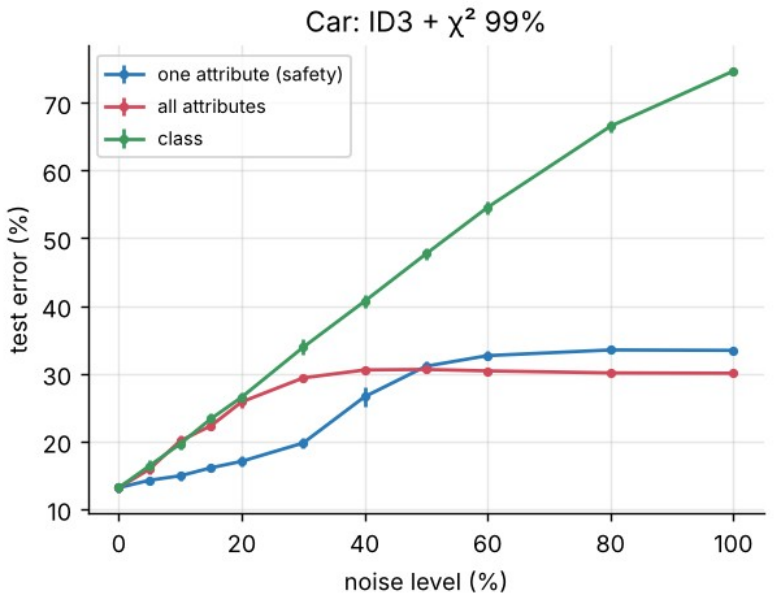

## How to read it

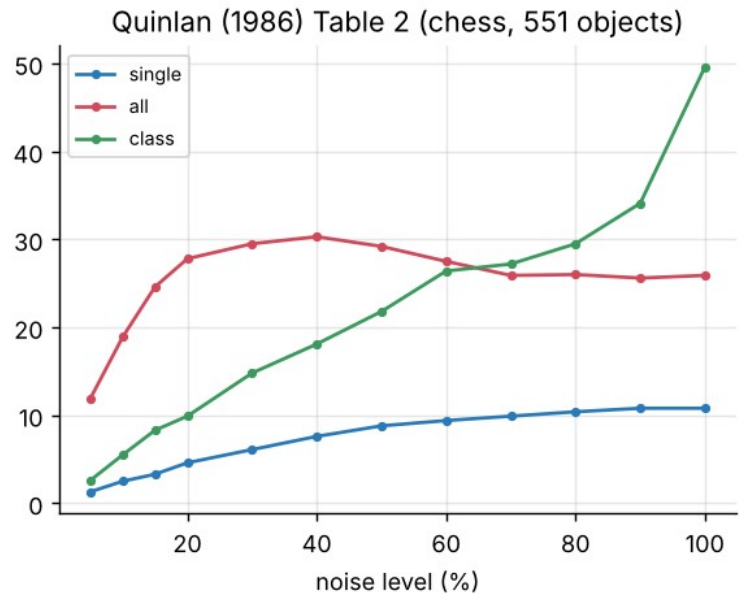
- Top: max_depth 0 … 8 and the full tree (train grey, test blue)
- Bottom: $\chi^2$ level α from 'none' to 0.0001; bars = tree size
- 20 stratified 70/30 splits, 95 % CI
- Ljubljana: best at depth 0, the majority class (69.8 %); the full tree has 231 nodes and 65.3 %

- **Clean concepts:** chess and car need deep trees (chess depth 3: 90.4 %, full: 99.3 %); a strict $\chi^2$ test costs car 8.6 points.
- **Noisy concepts:** votes is best at depth 1, one question about physician-fee-freeze (96.1 % vs 94.4 %, 3 nodes vs 34). Training accuracy keeps rising while test accuracy falls: overfitting.

# Noise on real data reproduces the shapes of the paper's Table 2



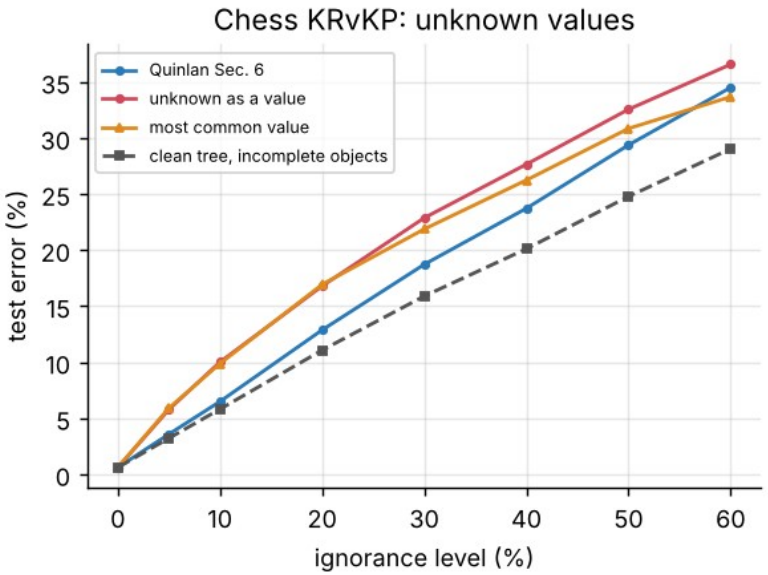



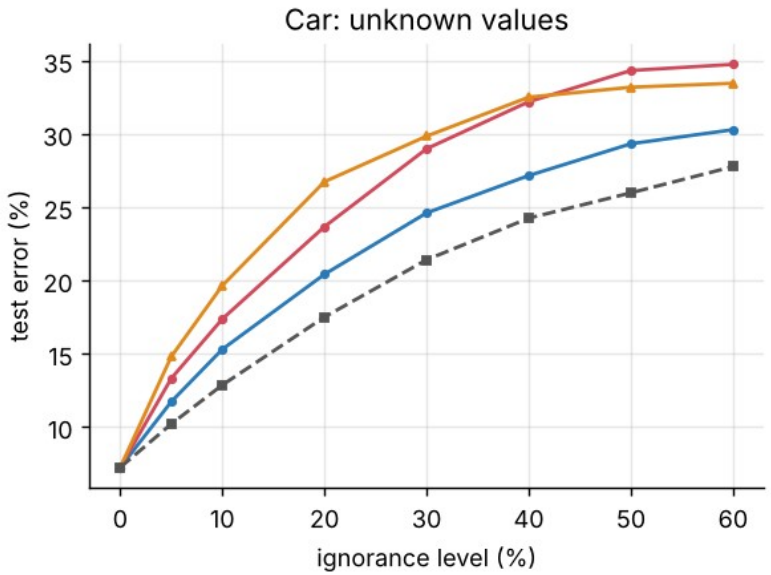

- **Class noise** grows almost linearly to random guessing: chess 49.8 % at 100 %, the paper 49.6 %.
- **Noise in all attributes** on car levels off at the majority-class error (30.1 %) once $\chi^2$ finds nothing relevant, as in the paper.
- **One attribute** (the root attribute) saturates: chess rises only to 8.8 %, the paper 10.8 %.

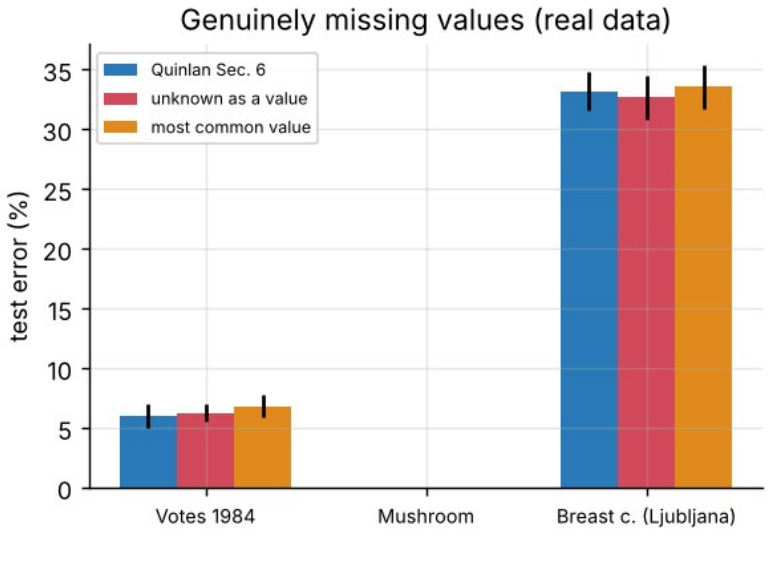
### Why $\chi^2$ matters (Section 5)

- Probabilistic leaf errs $2pn/(p+n)^2$; majority leaf $n/(p+n)$, smaller when $p > n$
- Ablation on chess with 20 % class noise: without $\chi^2$ 1160 nodes and 20.5 % error; with $\chi^2$ 53 nodes and 11.6 %

# Unknown values, gain bias and what we learned




- **Unknowns:** at 30 % ignorance on car, Quinlan's rule errs 24.6 % vs 29.0 % ('?' as a value) and 29.9 % (mode). On genuinely missing data the three tie.
- **Gain bias:** an irrelevant 150-valued id_code enters the gain tree (171 nodes, 95.5 %); the gain ratio never uses it (13 nodes, 100 %).
- **Windowing:** 304 of 2 237 chess objects gave a tree with zero training errors.

## Conclusions

- Code reproduces the paper's numbers and recovers a known tree
- Depth limits / $\chi^2$ help noisy data, hurt clean data
- Subset tests fix multi-way fragmentation
- github.com/Vinayak069/id3-decision-trees